EVENTUALLY SURJECTIVE MAPS GENERATED IN DEGREE 1 GIVE CLOSED IMMERSIONS

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Replacement by the image. lemma-eventual-iso-graded-rings-map-proj and lemma-morphism-proj-transitive (same chapter, not carded). Replace B by the image of A to B without changing X or the sheaves O_X(n).
- Surjective case. lemma-surjective-graded-rings-map-proj (carded this batch). Gives parts (1) and (2) and surjectivity in (3).
- Invertibility of the twists. lemma-apply-modules (same chapter, not carded). O_X(n) and O_Y(n) are invertible.

Source reference. Original lemma 01MT: sources/stacks-constructions/pinned-original-source.tex, source lines 2032–2032. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07ZE: sources/stacks-constructions/pinned-original-source.tex, source lines 2416–2416. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01MY: sources/stacks-constructions/pinned-original-source.tex, source lines 2271–2271. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01MZ: sources/stacks-constructions/pinned-original-source.tex, source lines 2355–2355. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01N0: sources/stacks-constructions/pinned-original-source.tex, source lines 2383–2383. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *With hypotheses and notation as in Lemma 01MY above. Assume $A_d \to B_d$ is surjective for $d \gg 0$ and that A is generated by A_1 over A_0. Then*

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) $U(\psi) = Y$,
(2) $r_\psi : Y \to X$ *is a closed immersion, and*
(3) *the maps* $\theta : r_\psi^* \mathcal{O}_X(n) \to \mathcal{O}_Y(n)$ *are isomorphisms.*

Proof. By Lemmas 07ZE and 01MZ we may replace B by the image of $A \to B$ without changing X or the sheaves $\mathcal{O}_X(n)$. Thus we may assume that $A \to B$ is surjective. By Lemma 01N0 we get (1) and (2) and surjectivity in (3). By Lemma 01MT we see that both $\mathcal{O}_X(n)$ and $\mathcal{O}_Y(n)$ are invertible. Hence θ is an isomorphism. □